

WBCAtt – Interpretable Multi-Task Network for White Blood Cell Classification and Morphology Tutoring

Alishba Zahid
Bachelors of Computer Engineering
Bsce22035
bsce22035@itu.edu.pk

Maheen Zahid
Bachelors of Computer Engineering
Bsce22018
bsce22018@itu.edu.pk

Syeda Maham Batool
Bachelors of Computer Engineering
Bsce22028
bsce22028@itu.edu.pk

Abstract—White blood cell (WBC) classification from microscopic blood smear images is a fundamental task in hematological diagnosis, supporting the detection of infections, immune disorders, and blood-related diseases such as leukemia. Manual examination by experts, while accurate, is time-consuming and prone to inter-observer variability. Recent advances in deep learning have enabled automated WBC classification with high accuracy; however, many studies lack reproducibility, comparative analysis under uniform settings, and practical deployment.

In this work, we reimplement a previously published deep learning-based WBC classification framework using the same four convolutional neural network (CNN) architectures reported in the original study. To extend the existing approach, we integrate EfficientNet-B3, a modern CNN architecture known for its compound scaling strategy and parameter efficiency. All models are trained and evaluated under identical preprocessing, training, and evaluation conditions to ensure fair comparison.

Furthermore, to bridge the gap between research and real-world application, we design and deploy a web-based user interface that allows users to select a trained model, upload a microscopic WBC image, and obtain real-time classification results. Experimental results demonstrate that EfficientNet-B3 achieves superior classification performance compared to the baseline models, while maintaining computational efficiency

Introduction

White blood cell (WBC) classification is essential for diagnosing hematological and immunological disorders such as leukemia and infections. Manual microscopic analysis, while accurate, is time-consuming and subject to observer variability. Recent deep learning approaches, particularly convolutional neural networks (CNNs), have achieved high accuracy in automated WBC classification; however, many studies lack reproducibility, fair model comparison, and practical deployment.

In this work, we reimplement an existing WBC

classification framework using four CNN architectures under identical experimental conditions and extend it by integrating EfficientNet-B3 to improve performance. Additionally, we develop a web-based interface that enables real-time model selection and image-based WBC classification, enhancing the practical applicability of the proposed system

Related Work

White Blood Cell Recognition. WBC recognition is crucial for the diagnosis of various diseases in hematology, leading to numerous studies focusing on the development of automatic WBC classifiers and the release of publicly available datasets. Prior to 2020, public WBC datasets were limited in size, containing only hundreds of images, which proved insufficient for leveraging state-of-the-art image classification models, such as Convolutional Neural Networks (CNNs). Recently, the introduction of larger datasets, containing around ten thousand images, has enabled subsequent studies to develop novel deep learning models that improve classification performance. However, despite the inherent requirement for explainability in medical applications, no prior datasets annotate morphological attributes, which are essential explanatory factors when hematologists recognize WBCs. As such, our work aims to address this significant yet overlooked aspect.

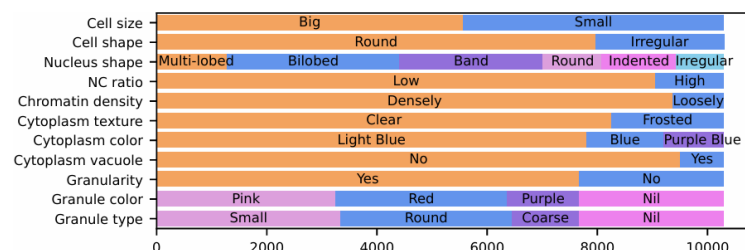
WBCAtt: WhiteBloodCell

Attribute Dataset

The attributes are categorized into four primary groups based on cell structure: overall cell, nucleus, cytoplasm, and granules, as described

in Sec. 3.1-3.4, respectively. The examples of these attributes are summarized in Figure 2. We utilized all images of typical WBCs from the PBC dataset [6], which encompassed 1,218 basophils, 3,117 eosinophils, 1,420 monocytes, 3,329 neutrophils, and 1,214 lymphocytes. We annotated 11 attributes for these 10,298 images, resulting in 113,278 image-attribute pairs, with the distribution for each attribute shown in Figure 3. Attribute Definition Process. Since there was no formally established set of attributes or ontology, we initiated our work with discussions with pathologists working in a laboratory at a healthcare company that develops digital cell imaging analyzers [28]. They provided five prominent attributes (see Coarse Morphological Attributes in Appendix) often used in identifying the five major WBC types, but also noted that the list was not exhaustive. Based on these initial keywords, we conducted a thorough review of relevant textbooks and research papers focusing on the morphological characteristics of WBCs. Subsequently, we refined the attribute set through further discussions with the pathologists and by manually inspecting approximately a thousand WBC images. This process yielded a total of 11 attributes, each supported by at least one medical literature reference. While the samples we inspected were the five major WBC types from healthy individuals, we expect the resulting attributes to sufficiently describe the morphological characteristics that may emerge in response to certain diseases, such as COVID-19 [29]. Moreover, these attributes provide pathologists with valuable insights into significant morphological abnormalities during microscopic examination, in accordance with the guidelines outlined in [30]. Annotation with Quality Control. To ensure reliable annotations across over 10k images, we devised a rigorous, iterative process involving pathologists, research scientists, and biomedical students who had strong knowledge of cell structures. In the initial stage, the students annotated the images, while being informed of the specific WBC category for each image. Following this, our research scientists meticulously examined each image and its assigned attributes, meaning that every image was inspected by at least two individuals. When ambiguities arose, we discussed with the

pathologists who defined the attributes with us, ensuring a consensus on labeling. Further details on the quality control are available in the Appendix. We assessed the reliability of our annotation process by replicating it on a subset of 1,000 images with different annotators. Out of the 11,000 attribute annotations, 10,569 were consistent with the original annotations, giving an agreement rate of approximately $10569/11000 \approx 96.1\%$. This high agreement rate demonstrates the robustness and reliability of our annotation process.



Dataset Description

The experiments are conducted on a publicly available white blood cell image dataset consisting of microscopic images representing multiple WBC categories. Each image corresponds to a single WBC instance captured under controlled staining and imaging conditions. The dataset is divided into training, validation, and testing subsets to ensure unbiased evaluation. All classes are preserved across splits to maintain balanced representation.

Data Preprocessing

Prior to training, all images undergo a standardized preprocessing pipeline. Images are resized to match the input resolution required by the selected CNN architectures. Pixel values are normalized using ImageNet statistics to facilitate transfer learning. To enhance model generalization and reduce overfitting, data augmentation techniques such as random rotation, horizontal flipping, and minor scaling are applied during training.

4.1 Baseline Models (Reimplemented)

To reproduce the original framework, four deep learning architectures are reimplemented: ResNet-50, VGG-16, Vision Transformer (ViT-B/16), and ConvNeXt. All models are initialized with ImageNet-pretrained weights and fine-tuned for white blood cell (WBC) classification by replacing the original classification head with a task-specific fully connected layer.

ResNet-50

ResNet-50 is a deep convolutional neural network that introduces residual connections to facilitate the training of very deep architectures by mitigating vanishing gradient issues. These skip connections allow the network to learn residual mappings, enabling effective extraction of discriminative features from complex visual patterns. Due to its robustness and strong representational capability, ResNet-50 has been widely adopted in medical image classification tasks. In this work, the pretrained ResNet-50 model is fine-tuned after replacing the final classification layer for WBC classification [1].

https://colab.research.google.com/drive/1UoHFk_smUQ0H0wFiko6zhypopWA6Bx5?usp=sharing

VGG-16

VGG-16 is a classical CNN architecture characterized by its simple and uniform design, consisting of stacked convolutional layers with small 3×3 kernels followed by max-pooling layers. Although computationally expensive, VGG-16 provides strong hierarchical feature representations and serves as a reliable baseline for performance comparison. The ImageNet-pretrained VGG-16 model is adapted by modifying its final fully connected layer to match the WBC classes [2].

https://colab.research.google.com/drive/1zyKDCcAjwgC5xudZRHDvc_GumOUQaV0U?usp=sharing

Vision Transformer (ViT-B/16)

The Vision Transformer (ViT-B/16) applies transformer-based self-attention mechanisms to image patches, enabling global context modeling across the entire image. Unlike convolutional networks, ViT captures long-range dependencies and global morphological patterns, which are beneficial for analyzing complex cell structures. In this study, the pretrained ViT-B/16 model is fine-tuned by replacing its classification head for WBC classification [3].

<https://colab.research.google.com/drive/1RwEjt8Pf9Co9YH0A04LvS74ToulvT49m?usp=sharing>

ConvNeXt

ConvNeXt is a modern convolutional architecture designed by incorporating architectural principles from vision transformers while retaining convolutional operations. It utilizes large kernel sizes, simplified layer structures, and optimized training strategies to achieve performance comparable to transformer-based models. ConvNeXt is included to evaluate the effectiveness of contemporary CNN designs for WBC classification and is fine-tuned using ImageNet-pretrained weights [4].

<https://colab.research.google.com/drive/1OLtVKfV7H4GupQUGZCHAMaDd3pf2MDvt?usp=sharing>

All baseline models are trained under identical preprocessing, optimization, and evaluation settings to ensure a fair and unbiased comparison.

Proposed Model: EfficientNet-B3

To further improve white blood cell (WBC) classification, the EfficientNet-B3 model is employed as the proposed architecture. EfficientNet is a family of convolutional neural networks designed to achieve a better trade-off between model accuracy and computational efficiency by optimizing network depth, width, and resolution simultaneously through a compound scaling method [5]. Among the variants, EfficientNet-B3 offers a strong balance of performance and resource requirements, making it suitable for medical image analysis tasks.

Architecture Overview

EfficientNet-B3 builds upon the Mobile Inverted Bottleneck Convolution (MBConv) blocks, which are optimized for feature extraction with depthwise separable convolutions. Each MBConv block consists of:

1. **Expansion layer** – increases the number of channels to capture more complex features.
2. **Depthwise convolution** – performs lightweight convolution over each channel separately.
3. **Squeeze-and-Excitation (SE) block** – recalibrates channel-wise feature responses for improved representation.
4. **Projection layer** – reduces the expanded channels back to the original size to maintain computational efficiency.

A key advantage of EfficientNet-B3 is its ability to capture fine-grained patterns while keeping the number of parameters and FLOPs lower than other high-capacity CNNs like ResNet-50 or VGG-16.

EfficientNet uses compound scaling to uniformly scale depth (d), width (w), and resolution (r) of the network:

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi$$

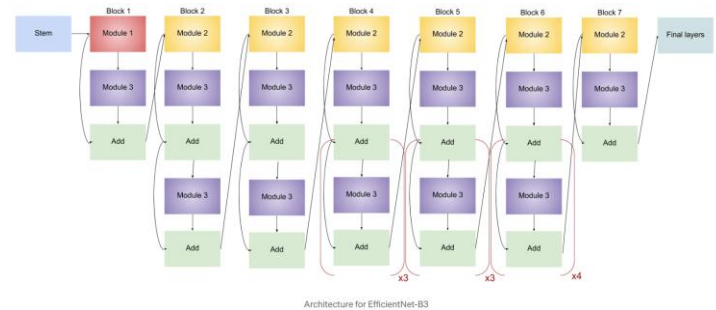
subject to the constraint:

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2, \quad \alpha, \beta, \gamma > 1$$

Here, ϕ is a user-defined compound coefficient that controls model scaling, while α , β , γ are constants determined through grid search. This strategy enables EfficientNet-B3 to increase capacity efficiently without overfitting or unnecessary computation.

EfficientNet-B3 benefits from **better feature extraction** due to its optimized depth, width, and resolution, allowing it to capture both local morphological details and global cellular patterns. Its efficient architecture also reduces training time and memory requirements compared to deeper networks like ResNet-50, while achieving competitive or superior accuracy.

<https://colab.research.google.com/drive/1wtqzThSuErx9AMiF-koRLJzTv2cIncF?usp=sharing>



Comparative Analysis of Baseline and Proposed Models

To evaluate the effectiveness of different deep learning architectures for white blood cell (WBC) classification, a comparative analysis of all five models is presented. The models—ResNet-50, VGG-16, Vision Transformer (ViT-B/16), ConvNeXt, and EfficientNet-B3—are compared in terms of key architectural characteristics, number of parameters, computational efficiency, feature extraction

capabilities, and suitability for medical imaging tasks. This comparison helps highlight the strengths and limitations of each model and justifies the choice of EfficientNet-B3 as the proposed model.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1-score
VGG	0.9961	0.9949	0.9964	0.9956
ResNet-50	0.9961	0.9951	0.9964	0.9957
Vision Transformer (ViT)	0.9961	0.9956	0.9976	0.9966
ConvNet	0.9971	0.9964	0.9970	0.9967
EfficientNet-B3	0.9971	0.9964	0.9970	0.9967

ATTRIBUTE PERFORMANCE
PER CELL TYPE

Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
VGG16	CellSize	Big	85.17	84.98	85.06
VGG16	CellSize	Small	81.73	81.91	81.81
VGG16	CellShape	Irregular	82.37	85.71	83.89
VGG16	CellShape	Round	95.79	94.54	95.15
VGG16	NucleusShape	Irregular	56.35	67.55	61.13
VGG16	NucleusShape	Segmented-Bilobed	74.98	74.42	74.70
VGG16	NucleusShape	Segmented-Multilobed	72.13	55.82	62.72
VGG16	NucleusShape	Unsegmented-Band	80.65	88.86	84.55
VGG16	NucleusShape	Unsegmented-Indented	81.87	76.22	78.91
VGG16	NucleusShape	Unsegmented-Round	83.31	82.19	82.60
VGG16	NCRatio	High	94.80	90.93	92.79
VGG16	NCRatio	Low	98.76	99.30	99.03
VGG16	ChromatinDensity	Densely	97.61	96.66	97.13

Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
VGG16	ChromatinDensity	Loosely	70.34	76.77	73.23
VGG16	CytoplasmVacuole	No	97.72	98.87	98.29
VGG16	CytoplasmVacuole	Yes	84.77	73.06	78.43
VGG16	CytoplasmTexture	Clear	98.52	95.99	97.23
VGG16	CytoplasmTexture	Frosted	86.05	94.43	89.98
VGG16	CytoplasmColor	Blue	80.39	77.48	78.83
VGG16	CytoplasmColor	LightBlue	98.90	99.11	99.00
VGG16	CytoplasmColor	PurpleBlue	73.27	75.20	74.01
VGG16	GranuleType	Coarse	98.76	99.52	99.14
VGG16	GranuleType	Nil	99.22	99.51	99.36
VGG16	GranuleType	Round	99.75	99.46	99.60
VGG16	GranuleType	Small	99.41	99.18	99.29
VGG16	GranuleColor	Nil	99.18	99.63	99.40
VGG16	GranuleColor	Pink	98.46	98.92	98.69
VGG16	GranuleColor	Purple	97.40	96.17	96.77
VGG16	GranuleColor	Red	99.82	99.42	99.62
VGG16	Granularity	No	99.34	99.42	99.38
VGG16	Granularity	Yes	99.80	99.77	99.78
Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
ResNet50	CellSize	Big	84.23	87.54	85.79
ResNet50	CellSize	Small	84.18	79.83	81.84
ResNet50	CellShape	Irregular	85.70	85.48	85.54
ResNet50	CellShape	Round	95.77	95.80	95.78
ResNet50	NucleusShape	Irregular	62.24	62.37	62.21
ResNet50	NucleusShape	Segmented-Bilobed	74.07	80.30	76.86

Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
ResNet50	NucleusShape	Segmented-Multilobed	71.43	59.97	64.95
ResNet50	NucleusShape	Unsegmented-Band	85.42	85.05	85.10
ResNet50	NucleusShape	Unsegmented-Indented	85.00	80.50	82.58
ResNet50	NucleusShape	Unsegmented-Round	84.35	86.01	85.07
ResNet50	NCRatio	High	96.16	91.11	93.56
ResNet50	NCRatio	Low	98.79	99.50	99.14
ResNet50	ChromatinDensity	Densely	97.97	96.67	97.31
ResNet50	ChromatinDensity	Loosely	71.14	80.37	75.46
ResNet50	CytoplasmVacuole	No	97.89	99.05	98.47
ResNet50	CytoplasmVacuole	Yes	87.53	75.10	80.67
ResNet50	CytoplasmTexture	Clear	98.78	96.49	97.62
ResNet50	CytoplasmTexture	Frosted	87.66	95.42	91.36
ResNet50	CytoplasmColor	Blue	83.30	87.53	85.19
ResNet50	CytoplasmColor	LightBlue	99.37	99.09	99.23
ResNet50	CytoplasmColor	PurpleBlue	82.07	77.55	79.54
ResNet50	GranuleType	Coarse	98.67	99.52	99.09
ResNet50	GranuleType	Nil	99.22	99.63	99.43
ResNet50	GranuleType	Round	99.82	99.89	99.86
ResNet50	GranuleType	Small	99.74	99.05	99.39
ResNet50	GranuleColor	Nil	99.34	99.71	99.53
ResNet50	GranuleColor	Pink	98.66	98.89	98.77
ResNet50	GranuleColor	Purple	97.82	95.99	96.90
ResNet50	GranuleColor	Red	99.75	99.93	99.84

Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
ResNet50	Granularity	No	99.38	99.46	99.42
Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
ViT-B/16	CellSize	Basophil	0.880	0.880	0.875862
ViT-B/16	CellShape	Basophil	0.888	0.888	0.842350
ViT-B/16	NucleusShape	Basophil	0.544	0.544	0.520330
ViT-B/16	NCRatio	Basophil	0.992	0.992	0.988016
ViT-B/16	ChromatinDensity	Basophil	1.000	1.000	1.000
ViT-B/16	CytoplasmVacuole	Basophil	1.000	1.000	1.000
ViT-B/16	CytoplasmTexture	Basophil	1.000	1.000	1.000
ViT-B/16	CytoplasmColor	Basophil	1.000	1.000	1.000
ViT-B/16	GranuleType	Basophil	0.992	0.992	0.988016
ViT-B/16	GranuleColor	Basophil	1.000	1.000	1.000
ViT-B/16	Granularity	Basophil	1.000	1.000	1.000
ViT-B/16	CellSize	Eosinophil	0.754777	0.754777	0.738895
ViT-B/16	CellShape	Eosinophil	0.952229	0.952229	0.948547
ViT-B/16	NucleusShape	Eosinophil	0.745223	0.745223	0.676386
ViT-B/16	NCRatio	Eosinophil	1.000	1.000	1.000
ViT-B/16	ChromatinDensity	Eosinophil	0.984076	0.984076	0.979338
ViT-B/16	CytoplasmVacuole	Eosinophil	0.993631	0.993631	0.990456
ViT-B/16	CytoplasmTexture	Eosinophil	1.000	1.000	1.000
ViT-B/16	CytoplasmColor	Eosinophil	1.000	1.000	1.000
ViT-B/16	GranuleType	Eosinophil	0.996815	0.996815	0.995225
ViT-B/16	GranuleColor	Eosinophil	1.000	1.000	1.000
ViT-B/16	Granularity	Eosinophil	1.000	1.000	1.000

Backbone	AttributeName	AttributeValue	Precision	Recall	F-measure
ViT-B/16	CellSize	Lymphocyte	0.990991	0.990991	0.986507
ViT-B/16	CellShape	Lymphocyte	0.873874	0.873874	0.866839
ViT-B/16	NucleusShape	Lymphocyte	0.846847	0.846847	0.825857
ViT-B/16	NCRatio	Lymphocyte	0.972973	0.972973	0.967370
ViT-B/16	ChromatinDensity	Lymphocyte	0.981982	0.981982	0.973055
ViT-B/16	CytoplasmVacuole	Lymphocyte	1.000	1.000	1.000
ViT-B/16	CytoplasmTexture	Lymphocyte	0.792793	0.792793	0.770188
ViT-B/16	CytoplasmColor	Lymphocyte	0.792793	0.792793	0.778378
ViT-B/16	GranuleType	Lymphocyte	1.000	1.000	1.000
ViT-B/16	GranuleColor	Lymphocyte	1.000	1.000	1.000
ViT-B/16	Granularity	Lymphocyte	1.000	1.000	1.000
ViT-B/16	CellSize	Monocyte	0.897260	0.897260	0.894295
ViT-B/16	CellShape	Monocyte	0.952055	0.952055	0.952053
ViT-B/16	NucleusShape	Monocyte	0.821918	0.821918	0.777812
ViT-B/16	NCRatio	Monocyte	0.938356	0.938356	0.941708
ViT-B/16	ChromatinDensity	Monocyte	0.732877	0.732877	0.670034
ViT-B/16	CytoplasmVacuole	Monocyte	0.938356	0.938356	0.938605
ViT-B/16	CytoplasmTexture	Monocyte	0.904110	0.904110	0.879223
ViT-B/16	CytoplasmColor	Monocyte	0.746575	0.746575	0.727968
ViT-B/16	GranuleType	Monocyte	1.000	1.000	1.000
ViT-B/16	GranuleColor	Monocyte	1.000	1.000	1.000
ViT-B/16	Granularity	Monocyte	1.000	1.000	1.000
ViT-B/16	CellSize	Neutrophil	0.817365	0.817365	0.816780
ViT-B/16	CellShape	Neutrophil	0.889222	0.889222	0.885705

Backbone	AttributeName	AttributeValue	Precision	Recall	F-measure
ViT-B/16	NucleusShape	Neutrophil	0.778443	0.778443	0.763592
ViT-B/16	NCRatio	Neutrophil	1.000	1.000	1.000
ViT-B/16	ChromatinDensity	Neutrophil	0.994012	0.994012	0.991027
ViT-B/16	CytoplasmVacuole	Neutrophil	0.961078	0.961078	0.951215
ViT-B/16	CytoplasmTexture	Neutrophil	1.000	1.000	1.000
ViT-B/16	CytoplasmColor	Neutrophil	1.000	1.000	1.000
ViT-B/16	GranuleType	Neutrophil	0.991018	0.991018	0.992994
ViT-B/16	GranuleColor	Neutrophil	0.973054	0.973054	0.971438
ViT-B/16	Granularity	Neutrophil	1.000	1.000	1.000
Backbone	AttributeName	AttributeValue	Precision	Recall	F-measure
ConvNeXt	CellSize	Basophil	0.867520	0.868565	0.867520
ConvNeXt	CellShape	Basophil	0.882727	0.868371	0.882727
ConvNeXt	NucleusShape	Basophil	0.612945	0.405607	0.612945
ConvNeXt	NCRatio	Basophil	0.984064	0.988016	0.984064
ConvNeXt	ChromatinDensity	Basophil	1.000000	1.000000	1.000000
ConvNeXt	CytoplasmVacuole	Basophil	1.000000	1.000000	1.000000
ConvNeXt	CytoplasmTexture	Basophil	1.000000	1.000000	1.000000
ConvNeXt	CytoplasmColor	Basophil	1.000000	1.000000	1.000000
ConvNeXt	GranuleType	Basophil	0.984064	0.988016	0.984064
ConvNeXt	GranuleColor	Basophil	1.000000	1.000000	1.000000
ConvNeXt	Granularity	Basophil	1.000000	1.000000	1.000000
ConvNeXt	CellSize	Eosinophil	0.779256	0.738854	0.779256
ConvNeXt	CellShape	Eosinophil	0.955087	0.950527	0.955087
ConvNeXt	NucleusShape	Eosinophil	0.655194	0.689085	0.655194

Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
ConvNet	NCRatio	Eosinophil	1.000000	1.000000	1.000000
ConvNet	ChromatinDensity	Eosinophil	0.974685	0.980933	0.974685
ConvNet	CytoplasmVacuole	Eosinophil	0.987302	0.990456	0.987302
ConvNet	CytoplasmTexture	Eosinophil	1.000000	1.000000	1.000000
ConvNet	CytoplasmColor	Eosinophil	1.000000	1.000000	1.000000
ConvNet	GranuleType	Eosinophil	0.993641	0.995225	0.993641
ConvNet	GranuleColor	Eosinophil	1.000000	1.000000	1.000000
ConvNet	Granularity	Eosinophil	1.000000	1.000000	1.000000
ConvNet	CellSize	Lymphocyte	0.982063	0.986507	0.982063
ConvNet	CellShape	Lymphocyte	0.907993	0.897122	0.907993
ConvNet	NucleusShape	Lymphocyte	0.875307	0.836634	0.875307
ConvNet	NCRatio	Lymphocyte	0.958458	0.959628	0.958458
ConvNet	ChromatinDensity	Lymphocyte	0.964289	0.973055	0.964289
ConvNet	CytoplasmVacuole	Lymphocyte	1.000000	1.000000	1.000000
ConvNet	CytoplasmTexture	Lymphocyte	0.748259	0.718846	0.748259
ConvNet	CytoplasmColor	Lymphocyte	0.746708	0.741718	0.746708
ConvNet	GranuleType	Lymphocyte	1.000000	1.000000	1.000000
ConvNet	GranuleColor	Lymphocyte	1.000000	1.000000	1.000000
ConvNet	Granularity	Lymphocyte	1.000000	1.000000	1.000000
ConvNet	CellSize	Monocyte	0.863014	0.863014	0.863014
ConvNet	CellShape	Monocyte	0.946546	0.945164	0.946546
ConvNet	NucleusShape	Monocyte	0.864726	0.830594	0.864726
ConvNet	NCRatio	Monocyte	0.935868	0.937002	0.935868
ConvNet	ChromatinDensity	Monocyte	0.772936	0.773417	0.772936

Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
ConvNet	CytoplasmVacuole	Monocyte	0.849315	0.849315	0.849315
ConvNet	CytoplasmTexture	Monocyte	0.892167	0.883932	0.892167
ConvNet	CytoplasmColor	Monocyte	0.725271	0.729017	0.725271
ConvNet	GranuleType	Monocyte	1.000000	1.000000	1.000000
ConvNet	GranuleColor	Monocyte	1.000000	1.000000	1.000000
ConvNet	Granularity	Monocyte	1.000000	1.000000	1.000000
ConvNet	CellSize	Neutrophil	0.802518	0.769284	0.802518
ConvNet	CellShape	Neutrophil	0.881254	0.835975	0.881254
ConvNet	NucleusShape	Neutrophil	0.725678	0.696749	0.725678
ConvNet	NCRatio	Neutrophil	1.000000	1.000000	1.000000
ConvNet	ChromatinDensity	Neutrophil	0.988060	0.991027	0.988060
ConvNet	CytoplasmVacuole	Neutrophil	0.959851	0.942235	0.959851
ConvNet	CytoplasmTexture	Neutrophil	1.000000	1.000000	1.000000
ConvNet	CytoplasmColor	Neutrophil	1.000000	1.000000	1.000000
ConvNet	GranuleType	Neutrophil	1.000000	1.000000	1.000000
ConvNet	GranuleColor	Neutrophil	0.973867	0.972566	0.973867
ConvNet	Granularity	Neutrophil	1.000000	1.000000	1.000000
Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
EfficientNet-B3	CellSize	Basophil	0.884419	0.875935	0.884419
EfficientNet-B3	CellShape	Basophil	0.894609	0.897473	0.894609
EfficientNet-B3	NucleusShape	Basophil	0.417639	0.423062	0.417639
EfficientNet-B3	NCRatio	Basophil	0.984064	0.988016	0.984064
EfficientNet-B3	ChromatinDensity	Basophil	1.000000	1.000000	1.000000
EfficientNet-B3	CytoplasmVacuole	Basophil	1.000000	1.000000	1.000000

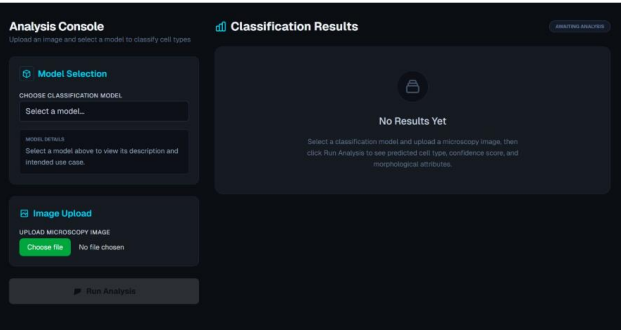
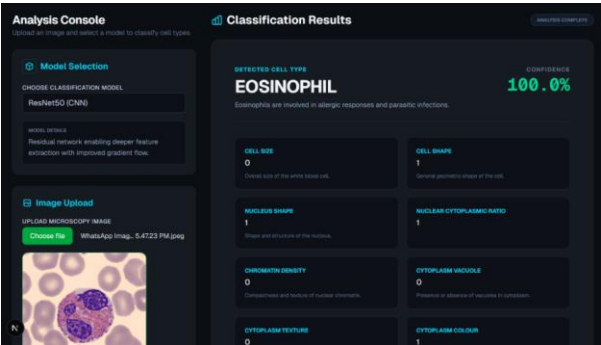
Backbone	AttributeName	Attribute Value	Precision	Recall	F-measure
EfficientNet-B3	CytoplasmTexture	Basophil	1.000000	1.000000	1.000000
EfficientNet-B3	CytoplasmColor	Basophil	1.000000	1.000000	1.000000
EfficientNet-B3	GranuleType	Basophil	0.984064	0.988016	0.984064
EfficientNet-B3	GranuleColor	Basophil	1.000000	1.000000	1.000000
EfficientNet-B3	Granularity	Basophil	1.000000	1.000000	1.000000
EfficientNet-B3	CellSize	Eosinophil	0.795801	0.785661	0.795801
EfficientNet-B3	CellShape	Eosinophil	0.973845	0.973733	0.973845
EfficientNet-B3	NucleusShape	Eosinophil	0.744452	0.693067	0.744452
EfficientNet-B3	NCRatio	Eosinophil	1.000000	1.000000	1.000000
EfficientNet-B3	ChromatinDensity	Eosinophil	0.984138	0.985158	0.984138
EfficientNet-B3	CytoplasmVacuole	Eosinophil	0.987302	0.990456	0.987302
EfficientNet-B3	CytoplasmTexture	Eosinophil	1.000000	1.000000	1.000000
EfficientNet-B3	CytoplasmColor	Eosinophil	1.000000	1.000000	1.000000
EfficientNet-B3	GranuleType	Eosinophil	0.993641	0.995225	0.993641
EfficientNet-B3	GranuleColor	Eosinophil	1.000000	1.000000	1.000000
EfficientNet-B3	Granularity	Eosinophil	1.000000	1.000000	1.000000
EfficientNet-B3	CellSize	Lymphocyte	0.982063	0.986507	0.982063
EfficientNet-B3	CellShape	Lymphocyte	0.900301	0.899862	0.900301
EfficientNet-B3	NucleusShape	Lymphocyte	0.743668	0.760456	0.743668
EfficientNet-B3	NCRatio	Lymphocyte	0.942433	0.945617	0.942433
EfficientNet-B3	ChromatinDensity	Lymphocyte	0.964289	0.973055	0.964289
EfficientNet-B3	CytoplasmVacuole	Lymphocyte	1.000000	1.000000	1.000000
EfficientNet-B3	CytoplasmTexture	Lymphocyte	0.772537	0.742273	0.772537
EfficientNet-B3	CytoplasmColor	Lymphocyte	0.705475	0.712793	0.705475

Backbone	AttributeName	Attribute Value	Precision	Recall	F-measure
EfficientNet-B3	GranuleType	Lymphocyte	1.000000	1.000000	1.000000
EfficientNet-B3	GranuleColor	Lymphocyte	1.000000	1.000000	1.000000
EfficientNet-B3	Granularity	Lymphocyte	1.000000	1.000000	1.000000
EfficientNet-B3	CellSize	Monocyte	0.899752	0.892664	0.899752
EfficientNet-B3	CellShape	Monocyte	0.931831	0.931494	0.931831
EfficientNet-B3	NucleusShape	Monocyte	0.761088	0.776591	0.761088
EfficientNet-B3	NCRatio	Monocyte	0.950201	0.951002	0.950201
EfficientNet-B3	ChromatinDensity	Monocyte	0.712921	0.648891	0.712921
EfficientNet-B3	CytoplasmVacuole	Monocyte	0.853819	0.843365	0.853819
EfficientNet-B3	CytoplasmTexture	Monocyte	0.858578	0.870105	0.858578
EfficientNet-B3	CytoplasmColor	Monocyte	0.638618	0.640565	0.638618
EfficientNet-B3	GranuleType	Monocyte	1.000000	1.000000	1.000000
EfficientNet-B3	GranuleColor	Monocyte	1.000000	1.000000	1.000000
EfficientNet-B3	Granularity	Monocyte	1.000000	1.000000	1.000000
EfficientNet-B3	CellSize	Neutrophil	0.861352	0.861680	0.861352
EfficientNet-B3	CellShape	Neutrophil	0.902028	0.902413	0.902028
EfficientNet-B3	NucleusShape	Neutrophil	0.744889	0.736688	0.744889
EfficientNet-B3	NCRatio	Neutrophil	1.000000	1.000000	1.000000
EfficientNet-B3	ChromatinDensity	Neutrophil	0.988042	0.989528	0.988042
EfficientNet-B3	CytoplasmVacuole	Neutrophil	0.959851	0.942235	0.959851
EfficientNet-B3	CytoplasmTexture	Neutrophil	1.000000	1.000000	1.000000
EfficientNet-B3	CytoplasmColor	Neutrophil	1.000000	1.000000	1.000000
EfficientNet-B3	GranuleType	Neutrophil	0.994012	0.994012	0.994012
EfficientNet-B3	GranuleColor	Neutrophil	0.970002	0.966732	0.970002

Backbone	Attribute Name	Attribute Value	Precision	Recall	F-measure
EfficientNet-B3	Granularity	Neutrophil	1.000000	0.998501	1.000000

User Interface and Model Integration

To provide an interactive platform for white blood cell (WBC) analysis, a user-friendly frontend interface was developed and seamlessly integrated with the trained deep learning models. The interface allows users to select one of the five available models—**VGG16**, **ResNet50**, **ViT-B/16**, **ConvNeXt**, or **EfficientNet-B3**—based on their preference or application requirements.



References

[1] X. Z. S. R. a. J. S. K. He, "Deep residual learning for image recognition," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[2] K. S. a. A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *International Conference on Learning Representations (ICLR)*, 2015.

[3] A. D. e. al., "An image is worth 16×16 words: Transformers for image recognition at scale," *International Conference on Learning Representations (ICLR)*, 2021.

[4] Z. L. e. al., "A ConvNet for the 2020s," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[5] M. T. a. Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019.

